Test Results for Forensic Media Preparation Tool:
Roadkil's Disk Wipe Version 1.2

Federated Testing Suite for Forensic Media Preparation

June 2020

Homeland Security

Science and Technology

This report was prepared for the Department of Homeland Security (DHS) Science and Technology Directorate (S&T) by the Office of Law Enforcement Standards of the National Institute of Standards and Technology.

For additional information about ongoing DHS S&T cybersecurity projects, please visit https://www.dhs.gov/science-and-technology/cybersecurity

June 2020

Test Results for Forensic Media Preparation Tool:
Roadkil's Disk Wipe Version 1.2

Federated Testing Suite for Forensic Media Preparation

Contents

Introduction 5
How to Read This Report 6
1. Tool Description 7
2. Testing Organization 7
3. Results Summary 7
4.Test Environment & Selected Test Configurations 7
4.1 Test Hardware and Software 7
4.2 Defined Test Configurations 8
4.3 Test Drive Information and Layouts 9
5. Test Results by Test Configuration 10
5.1 Results Summary 11
5.2 Test Result Details by Configuration 12
5.2.1 Test Result Details for Configuration 001 12
5.2.2 Test Result Details for Configuration 002 13
5.2.3 Test Result Details for Configuration 003 14
5.2.4 Test Result Details for Configuration 004 17
5.2.5 Test Result Details for Configuration 005 19
5.2.6 Test Result Details for Configuration 006 20
5.2.7 Test Result Details for Configuration 007 21
5.2.8 Test Result Details for Configuration 008 22
5.2.9 Test Result Details for Configuration 009 24
5.2.10 Test Result Details for Configuration 010 25
6. Appendix: Additional Details 27
6.1 Test Configuration Administrative Details 27

Introduction

The Computer Forensics Tool Testing (CFTT) program is a joint project of the Department of Homeland Security Science and Technology Directorate (DHS S&T), the National Institute of Justice, and the National Institute of Standards and Technology (NIST) Special Programs Office and Information Technology Laboratory. CFTT is supported by other organizations, including the Federal Bureau of Investigation, the U.S. Department of Defense Cyber Crime Center, U.S. Internal Revenue Service Criminal Investigation Division Electronic Crimes Program, and the DHS Bureau of Immigration and Customs Enforcement, U.S. Customs and Border Protection and U.S. Secret Service. The objective of the CFTT program is to provide measurable assurance to practitioners, researchers, and other applicable users that the tools used in computer forensics investigations provide accurate results. Accomplishing this requires the development of specifications and test methods for computer forensics tools and subsequent testing of specific tools against those specifications.

Test results provide the information necessary for developers to improve tools, users to make informed choices, and the legal community and others to understand the tools' capabilities. The CFTT approach to testing computer forensics tools is based on well-recognized methodologies for conformance and quality testing. Interested parties in the computer forensics community can review and comment on the specifications and test methods posted on the CFTT website (https://www.cftt.nist.gov/).

This document reports the results from testing the forensic media preparation function of Roadkil's Disk Wipe Version 1.2 using the CFTT Federated Testing Test Suite for Forensic Media Preparation, Version 5.

Federated Testing is an expansion of the CFTT program to provide forensic investigators and labs with test materials for tool testing and to support shared test reports. The goal of Federated Testing is to help forensic investigators to test the tools that they use in their labs and to enable sharing of tool test results. CFTT's Federated Testing Forensic Tool Testing Environment and included test suites can be downloaded from http://www.cftt.nist.gov/federated-testing.html and used to test forensic tools. The results can be optionally shared with CFTT, reviewed by CFTT staff, and then shared with the community.

Test results from other tools can be found on DHS's computer forensics webpage, https://www.dhs.gov/science-and-technology/nist-cftt-reports.

How to Read This Report

This report is organized into the following sections:

1. **Tool Description:** The tool name, version, and developer information are listed.
2. **Testing Organization:** Contact information and approvals.
3. **Results Summary:** This section identifies any significant anomalies observed in the test runs. This section provides a narrative of key findings identifying where the tool meets expectations and provides a summary of any ways the tool did not meet expectations. The section also provides any observations of interest about the tool or about testing the tool including any observed limitations or organization-imposed restrictions on tool use.
4. **Test Environment & Selected Test Configurations:** Description of hardware and software used in tool testing, the test drives used, and a list of the applicable test configurations from the Federated Testing Forensic Media Preparation Test Suite.
5. **Test Results by Test Configuration:** Automatically generated test results that identify anomalies.
6. **Appendix: Additional Details:** Additional administrative details for each test configuration such as, who ran the test, when the test was run, computer used, etc.

Test Results for Forensic Media Preparation Tool: Roadkil's Disk Wipe Version 1.2

1. Tool Description

Tool Name: Disk Wipe
Tool Version: 1.2
Operating System: Microsoft Windows 10 Pro Version 1903, build 18362.720

Vendor Contact:

Vendor name:	Roadkil
Web:	roadkil.net

2. Testing Organization

Organization conducting test: Nova Southeastern University
Contact: lh1490@mynsu.nova.edu
Report date: March 26, 2020
Authored by: Lazaro Herrera

This test report was generated using CFTT's Federated Testing Forensic Tool Testing Environment, see Federated Testing Home Page.

3. Results Summary

Disk Wipe was not able to fully wipe media. The tool appears to skip over the last few sectors of every drive tested.

4. Test Environment & Selected Test Configurations

This section describes the test hardware and software, test configurations, and test drives used in testing.

4.1 Test Hardware and Software

Hardware: ***Custom Ryzen 7 PC, 16GB RAM***

Operating system: ***Windows 10 Pro Version 1903, Build Number 18362.720.***

Software: ***FTK Imager (sector checks)***

4.2 Defined Test Configurations

The following table describes each defined configuration of test drive and wipe method.

The columns are as follows:

- **Config:** The test configuration ID.
- **Drive Type:** The drive size category and interface type.
- **Host Interface:** The type of connection used to connect the test drive to the test computer.
- **Connection:** Either *direct* or *bridge*. Indicates if the test drive was connected to the test computer directly or via a bridge. If connected via a bridge, the bridge description is included.
- **Hidden Sectors:** Indicates the presence and type of hidden sectors.
- **Wipe Method:** The selected method for wiping a drive.

•

Config	Drive Type	Host Interface	Connection	Hidden Sectors	Wipe Method
001	Small USB (any type) < 2TB	USB3	Direct	None	Overwrite
002	Small USB (any type) < 2TB	USB3	Direct	None	Overwrite
003	Small USB (any type) < 2TB	USB3	Bridge: SD card to USB stick	None	Overwrite
004	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite
005	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite
006	Big IDE/PATA > 140GB	USB3	Direct	None	Overwrite
007	SATA w/ NCQ support (modern drive)	USB3	Bridge: Coolgear	None	Overwrite
008	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite
009	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite
010	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite

Note on Serial Advanced Technology Attachment (SATA) w/ NCQ support vs SATA w/o NCQ support: forensic media preparation tools treat drives differently based on whether they support NCQ (Native Command Queuing). SATA Hard Disk Drive (HDD)s manufactured before 2005, and some older Solid State Drives (SSD)s do not support NCQ. Newer HDDs manufactured in 2005 or later support NCQ.

4.3 Test Drive Information and Layouts

The following table describes the test drive and its layout for each test configuration.

- **Config:** The test configuration ID.
- **Drive Type:** The drive size category and interface type.
- **Manufacturer/Model:** The drive manufacturer and model.
- **Drive Size:** The drive size in sectors and Mega/Giga bytes.
- **Hidden Sectors:** The size in sectors of any hidden area and the type of hidden area.

Config	Drive Type	Manufacturer/Model	Drive Size	Hidden Sectors
001	Small USB (any type) < 2TB	Nvidia 64GB Flash Drive	124,822,487 (59GB)	0
002	Small USB (any type) < 2TB	Nvidia 64GB Flash Drive (30 min cooldown)	124,822,487 (59GB)	0
003	Small USB (any type) < 2TB	Netac 8Gb SD Card (FTK Imager)	15,613,952 (7GB)	0
004	SATA w/ NCQ support (modern drive)	Inland 120Gb SSD	234,441,648 (111GB)	0
005	SATA w/ NCQ support (modern drive)	WD Green 120 Gb SSD	234,455,040 (111GB)	0
006	Big IDE/PATA > 140GB	Maxtor 160Gb HDD	312,500,000 (149GB)	0
007	SATA w/ NCQ support (modern drive)	Corsair Force 200Gb SSD	390,721,968 (186GB)	0
008	SATA w/ NCQ support (modern drive)	Seagate Momentus 320Gb	625,142,448 (298GB)	0
009	SATA w/ NCQ support (modern drive)	Toshiba 1TB HDD	1,953,525,168 (931GB)	0
010	SATA w/ NCQ support (modern drive)	WD Blue 1TB HDD	1,953,525,168 (931GB)	0

5. Test Results by Test Configuration

This section has two subsections: a summary of the test results and detailed results for each test configuration.

5.1 Results Summary

The following table reports the overall result for each tested configuration. An entry of *Anomaly* in the Results column means that some sectors were not wiped. An entry of *As Expected* in the Results column means that all sectors were completely overwritten or erased.

Config	Drive Type	Host Interface	Connection	Hidden Sectors	Wipe Method	Results
001	Small USB (any type) < 2TB	USB3	Direct	None	Overwrite	Anomaly
002	Small USB (any type) < 2TB	USB3	Direct	None	Overwrite	Anomaly
003	Small USB (any type) < 2TB	USB3	Bridge: SD card to USB stick	None	Overwrite	Anomaly
004	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite	Anomaly
005	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite	Anomaly
006	Big IDE/PATA > 140GB	USB3	Direct	None	Overwrite	Anomaly
007	SATA w/ NCQ support (modern drive)	USB3	Bridge: Coolgear	None	Overwrite	Anomaly
008	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite	Anomaly
009	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite	Anomaly
010	SATA w/ NCQ support (modern drive)	SATA	Direct	None	Overwrite	Anomaly

5.2 Test Result Details by Configuration

This section presents the detailed analysis of each test configuration. Each analysis is presented as a table of sector runs for sectors as identified as either *unchanged*, *overwritten*, or *shifted*. A successful test result is for all sectors to be overwritten.
The columns of the tables of sector runs are as follows:

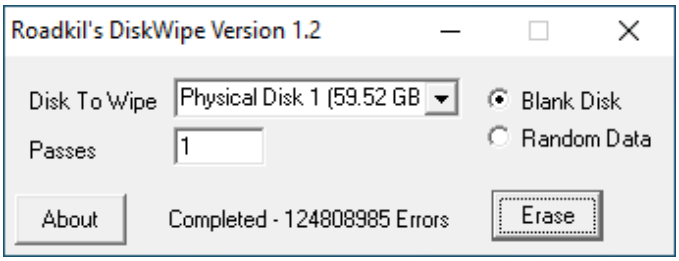
- **Result Type:** Category of result, either *overwritten* or *unchanged*. Sectors that have been relocated (still with original content) are classified as *shifted* and are considered as a variation on *unchanged*.
- **N Sectors:** The number of sectors in the category.
- **N Runs:** The number of sector runs in the category.
- **Start LBA:** For each sector run, this is the LBA of the first sector of the run.
- **End LBA:** For each sector run, this is the LBA of the last sector of the run.
- **Run Length:** For each sector run, the number of sectors in the run.

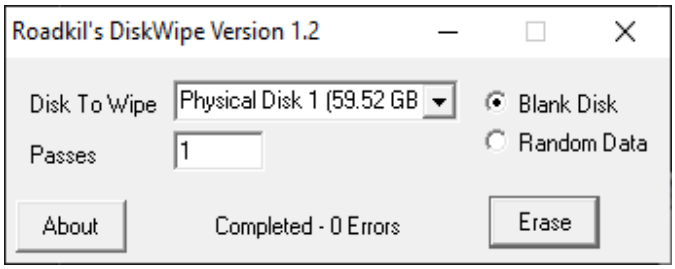
5.2.1 Test Result Details for Configuration 001

Anomaly Detected: Configuration 001, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**124,808,985**	**1**	**0**	**124,808,984**	**124,808,985**
unchanged	**13,502**	**1**	**124,808,985**	**124,822,486**	**13,502**

Media had to be cleaned using diskpart because tool would fail with sector errors





Tool failed to overwrite the last set of sectors for media.

5.2.2 Test Result Details for Configuration 002

Anomaly Detected: Configuration 002, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**124,808,985**	**1**	**0**	**124,808,984**	**124,808,985**
unchanged	**13,502**	**1**	**124,808,985**	**124,822,486**	**13,502**

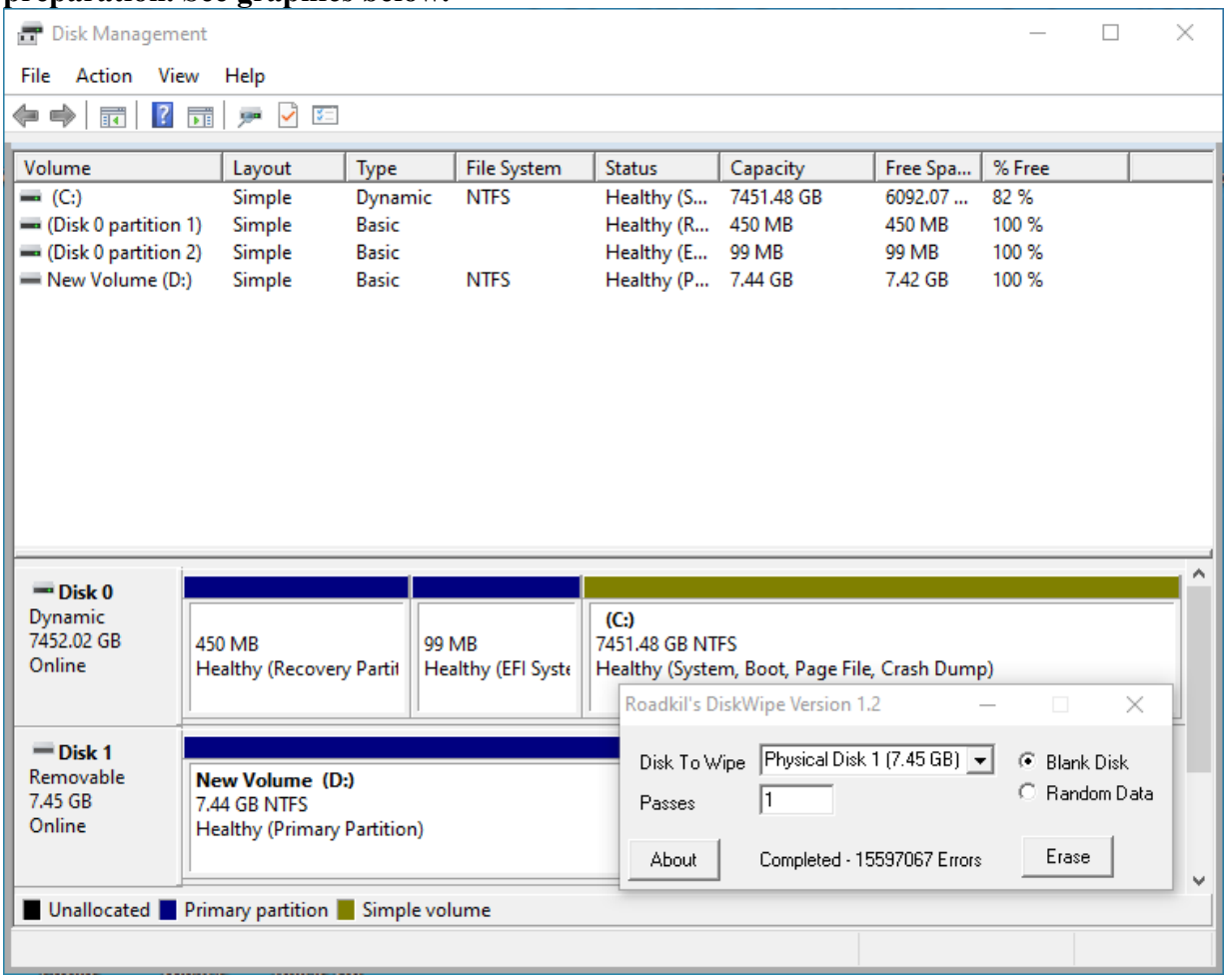
Tool failed to overwrite the last set of sectors for media.

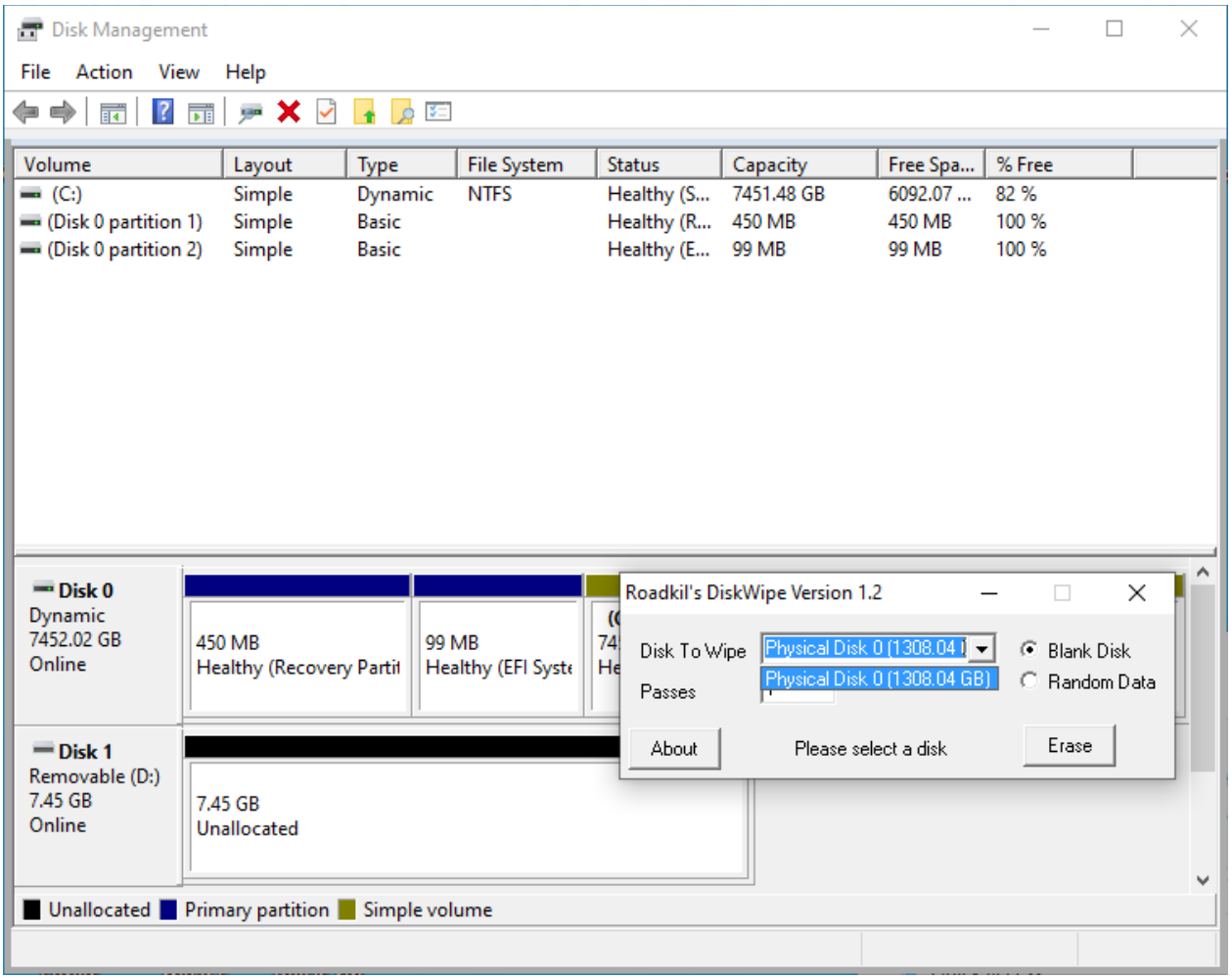
5.2.3 Test Result Details for Configuration 003

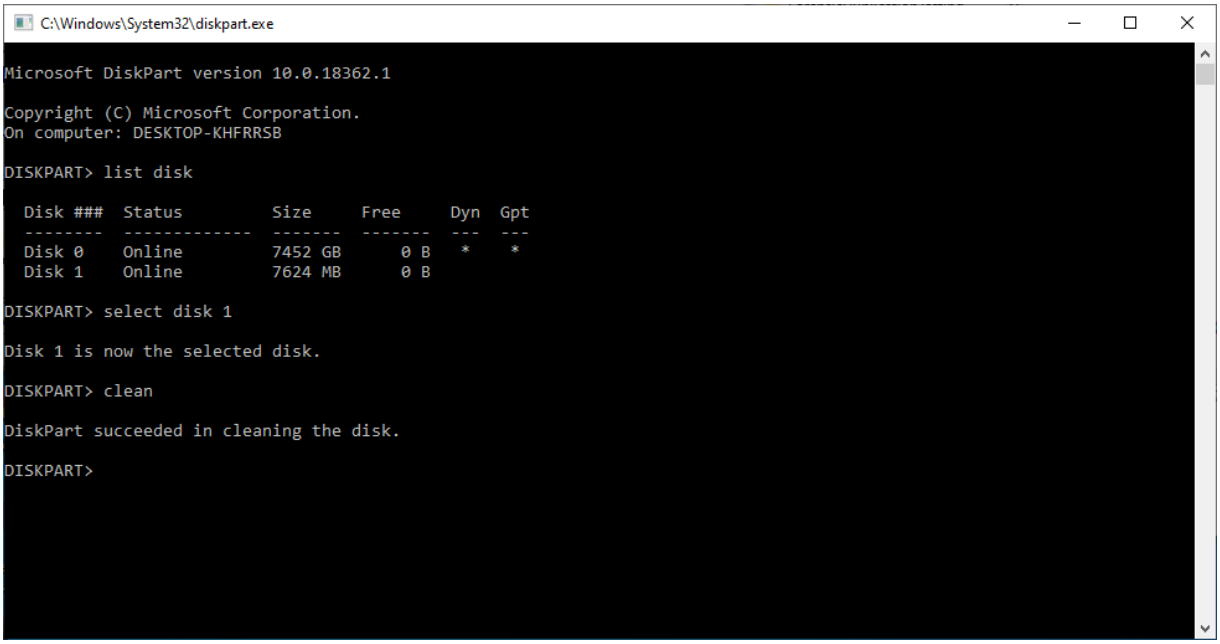
Anomaly Detected: Configuration 003, not all sectors overwritten

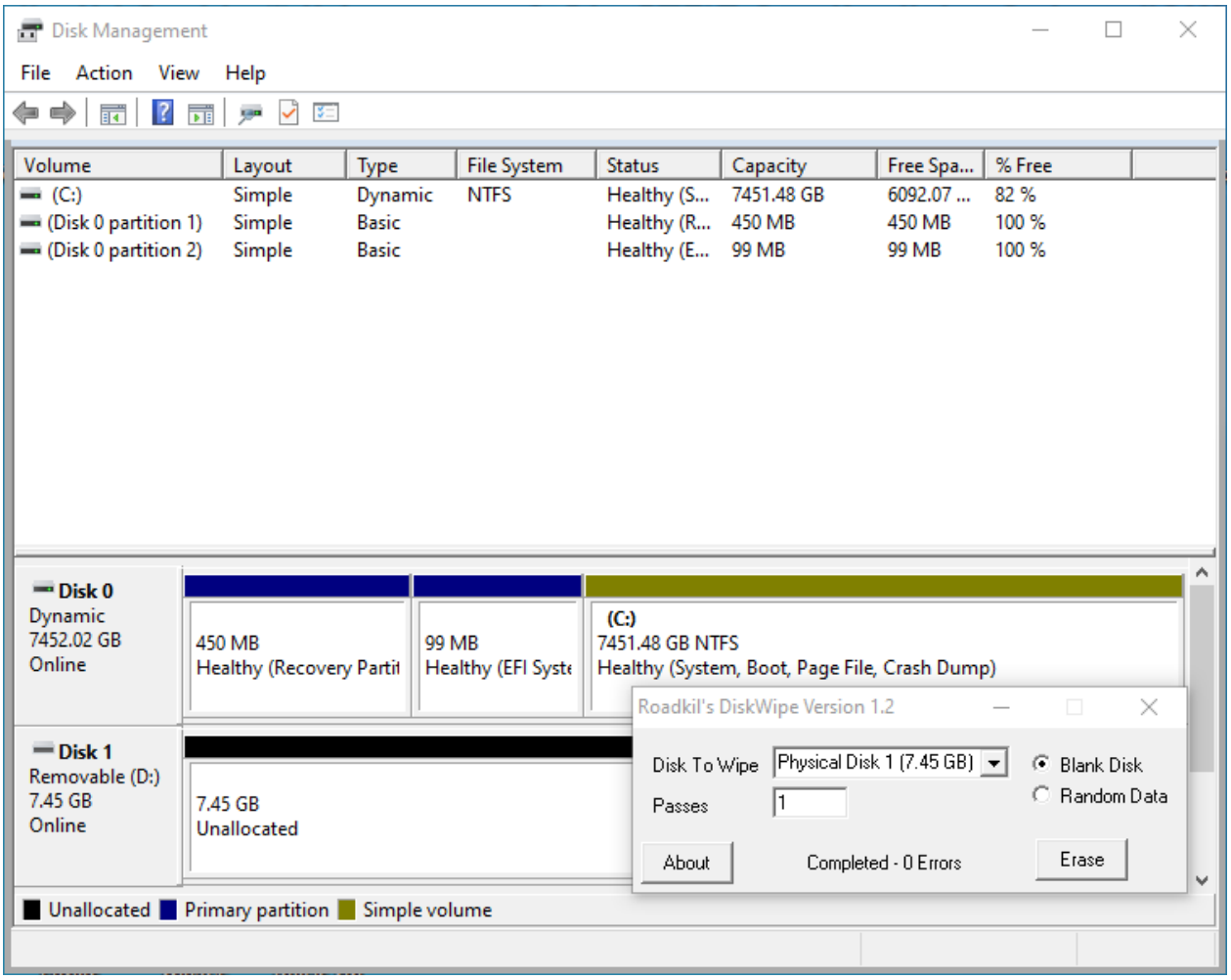
Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**15,601,163**	**2**	**0**	**15,599,114**	**15,599,115**
overwritten (hex fill)	**15,601,163**	**2**	**15,611,904**	**15,613,951**	**2,048**
unchanged	**12,788**	**1**	**15,599,115**	**15,611,902**	**12,788**
overwritten (non-hex fill)	**1**	**1**	**15,611,903**	**15,611,903**	**1**

Media had to be cleaned using diskpart before tool would run. Roadkil encountered issues wiping certain drives unless partitions were removed. These issues appeared before CFTT preparation withNew Technology File System (NTFS) partition) and after CFTT preparation. See graphics below.




Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Spa... % Free
(C:) Simple Dynamic NTFS Healthy (S... 7451.48 GB 6092.07 ... 82 %
(Disk 0 partition 1) Simple Basic Healthy (R... 450 MB 450 MB 100 %
(Disk 0 partition 2) Simple Basic Healthy (E... 99 MB 99 MB 100 %
Disk 0
Dynamic
7452.02 GB
Online
450 MB
Healthy (Recovery Partit
99 MB
Healthy (EFI Syste
Disk 1
Removable (D:)
7.45 GB
Online
7.45 GB
Unallocated
Roadkil's DiskWipe Version 1.2
Disk To Wipe
Physical Disk 0 (1308.04 I
Physical Disk 0 (1308.04 GB)
Passes
Blank Disk
Random Data
About
Please select a disk
Erase
Unallocated
Primary partition
Simple volume


C:\Windows\System32\diskpart.exe
Microsoft DiskPart version 10.0.18362.1
Copyright (C) Microsoft Corporation.
On computer: DESKTOP-KHFRRSB
DISKPART> list disk
Disk ### Status Size Free Dyn Gpt
Disk 0 Online 7452 GB 0 B * *
Disk 1 Online 7624 MB 0 B
DISKPART> select disk 1
Disk 1 is now the selected disk.
DISKPART> clean
DiskPart succeeded in cleaning the disk.
DISKPART>

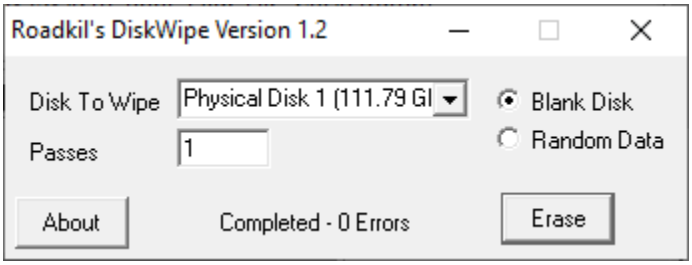


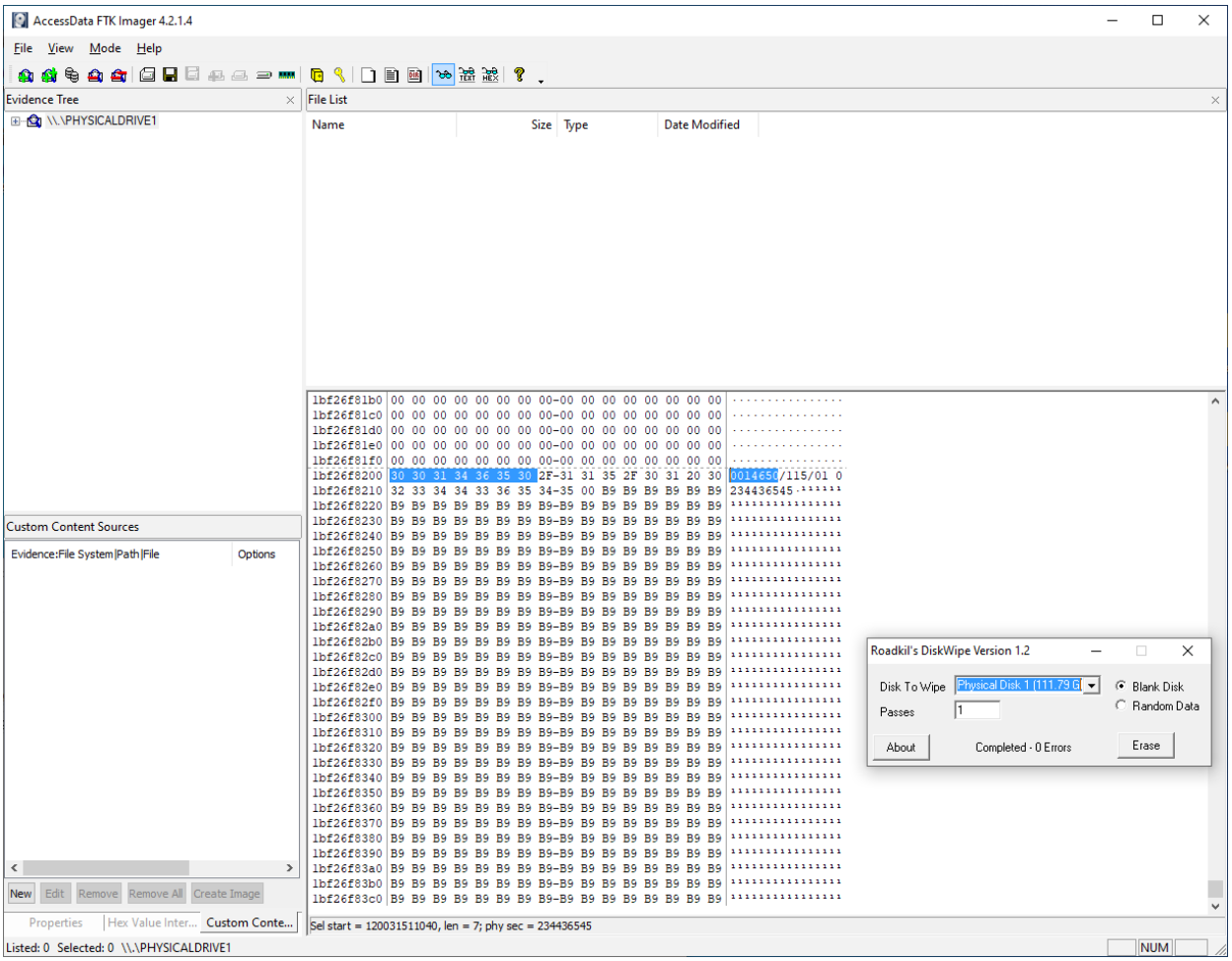
Tool failed to overwrite the last set of sectors for media.

5.2.4 Test Result Details for Configuration 004

Anomaly Detected: Configuration 004, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**234,436,545**	**1**	**0**	**234,436,544**	**234,436,545**
unchanged	**5,103**	**1**	**234,436,545**	**234,441,647**	**5,103**



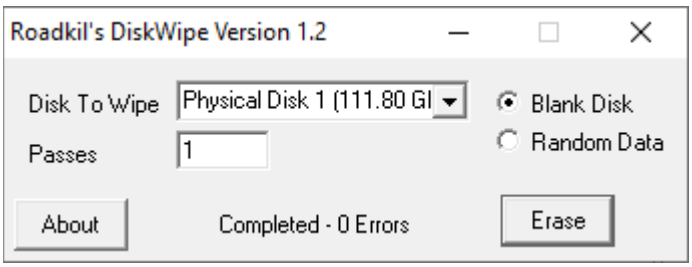


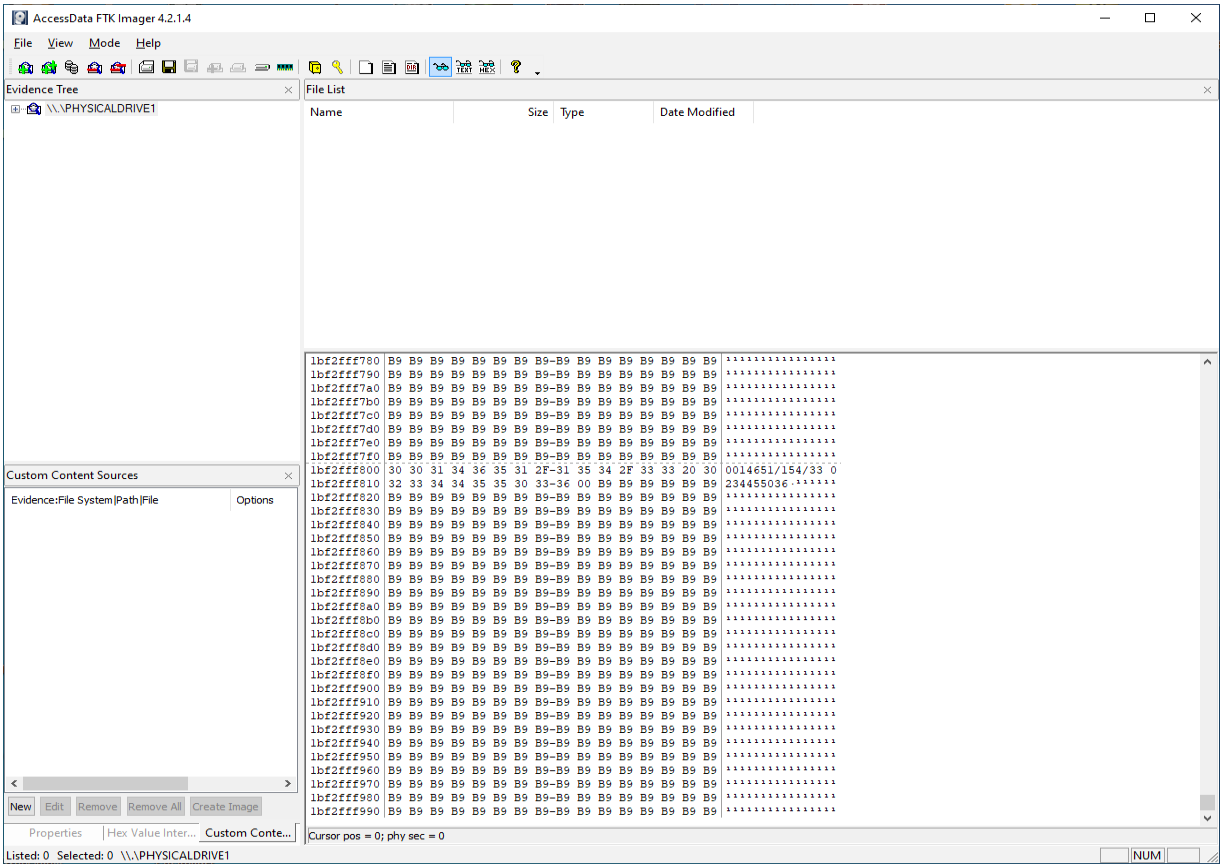
Tool failed to overwrite the last set of sectors for media.

5.2.5 Test Result Details for Configuration 005

Anomaly Detected: Configuration 005, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**234,452,610**	**1**	**0**	**234,452,609**	**234,452,610**
unchanged	**2,430**	**1**	**234,452,610**	**234,455,039**	**2,430**



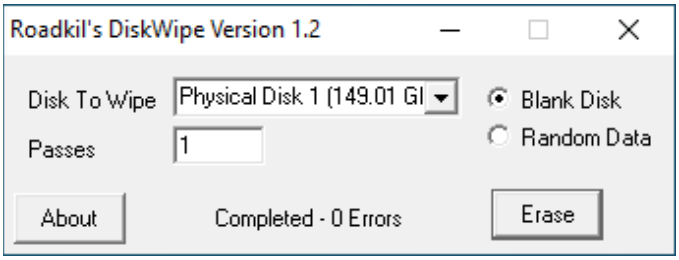


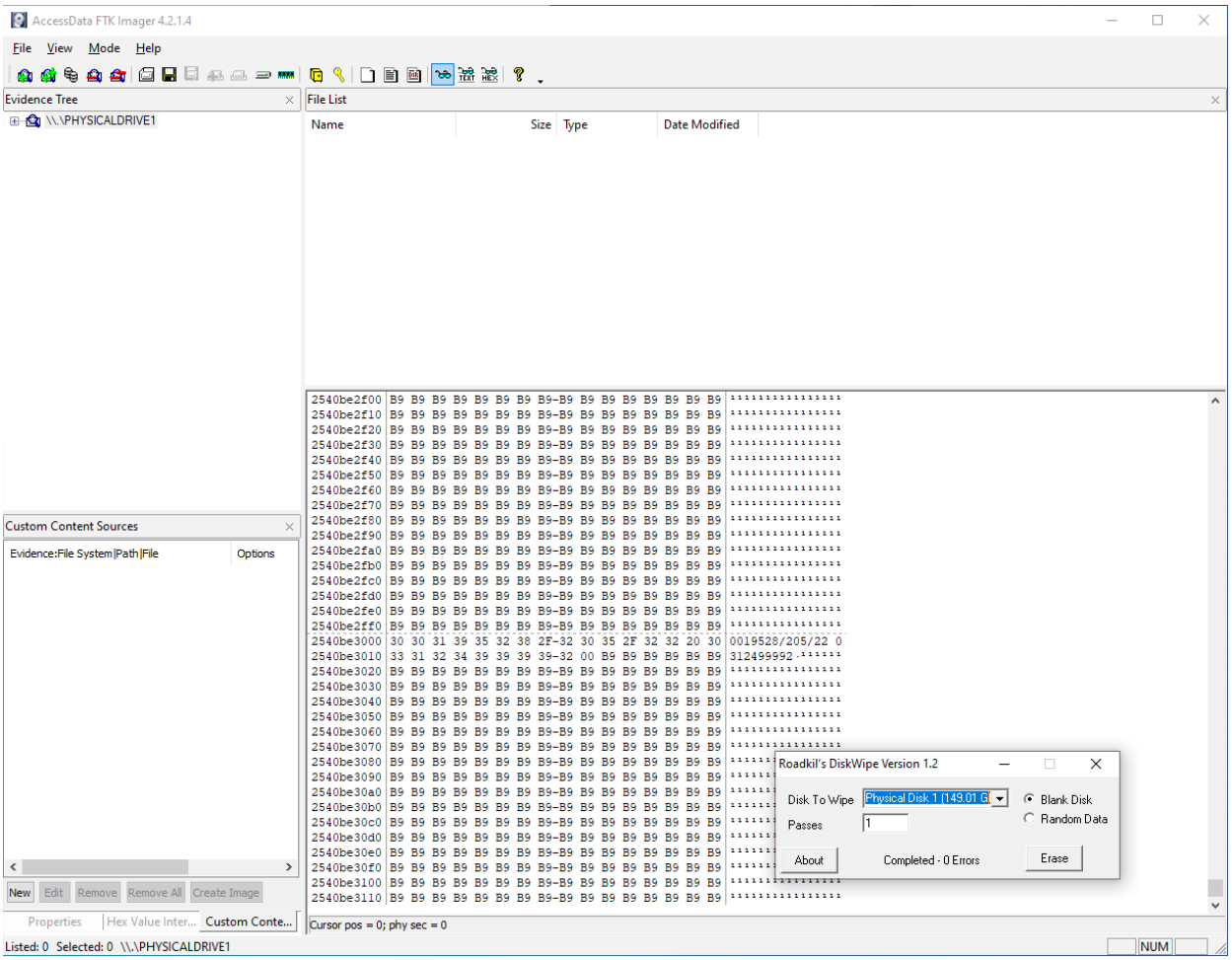
Tool failed to overwrite the last set of sectors for media.

5.2.6 Test Result Details for Configuration 006

Anomaly Detected: Configuration 006, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**312,496,380**	**1**	**0**	**312,496,379**	**312,496,380**
unchanged	**3,620**	**1**	**312,496,380**	**312,499,999**	**3,620**



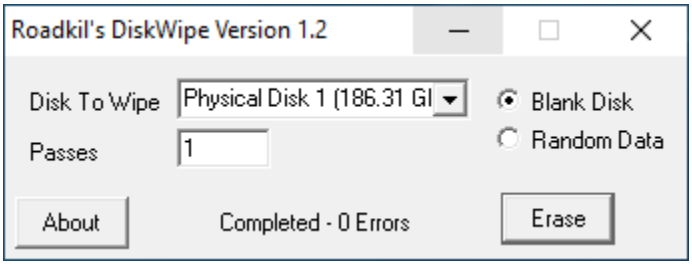


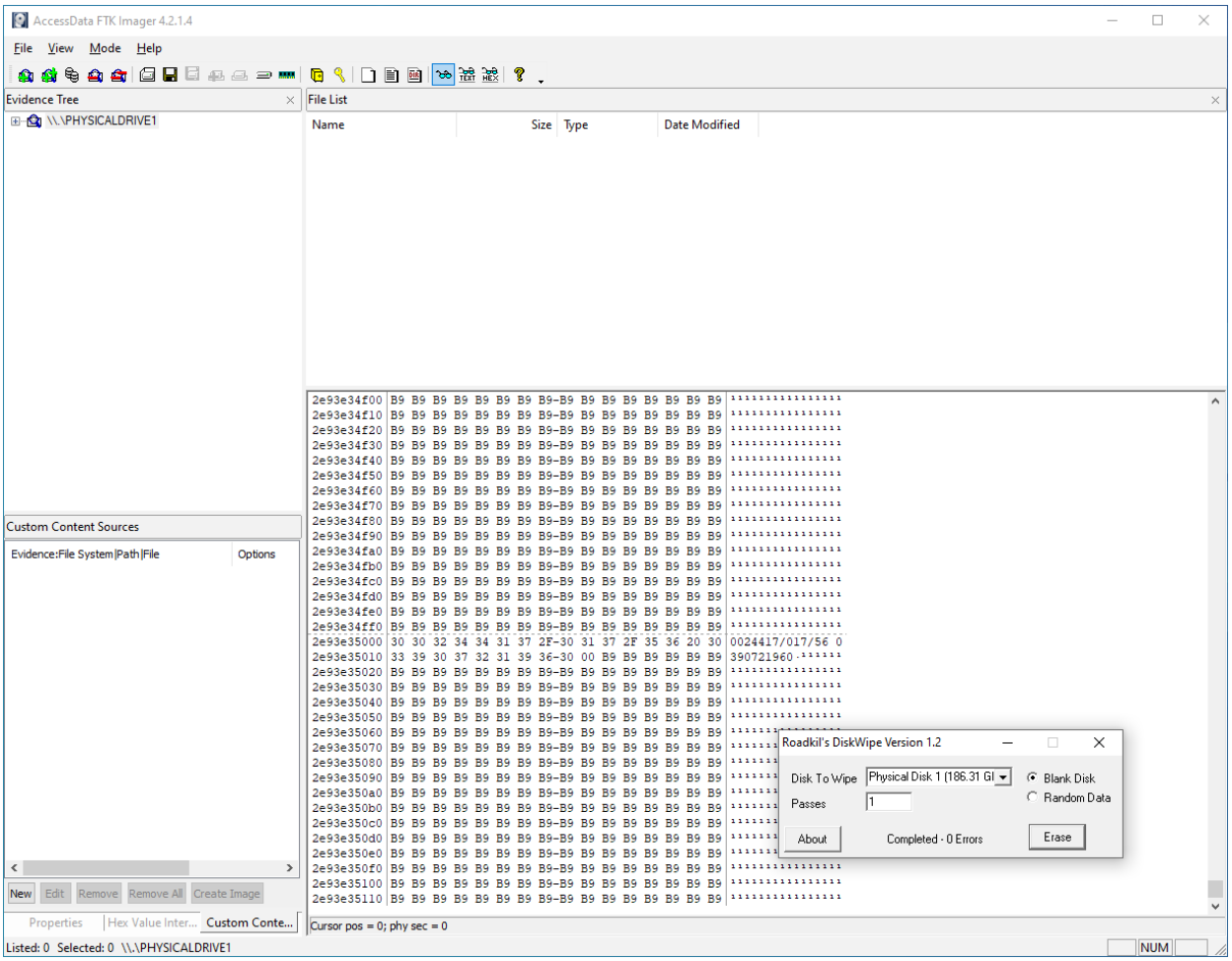
Tool failed to overwrite the last set of sectors for media.

5.2.7 Test Result Details for Configuration 007

Anomaly Detected: Configuration 007, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**390,716,865**	**1**	**0**	**390,716,864**	**390,716,865**
unchanged	**5,103**	**1**	**390,716,865**	**390,721,967**	**5,103**



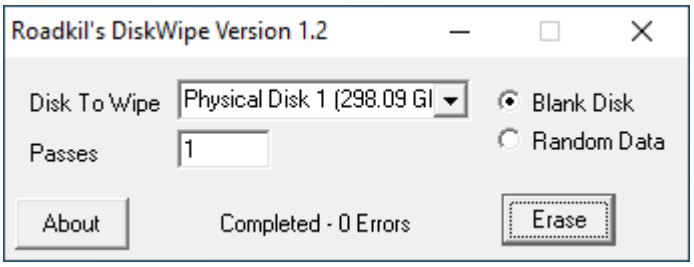


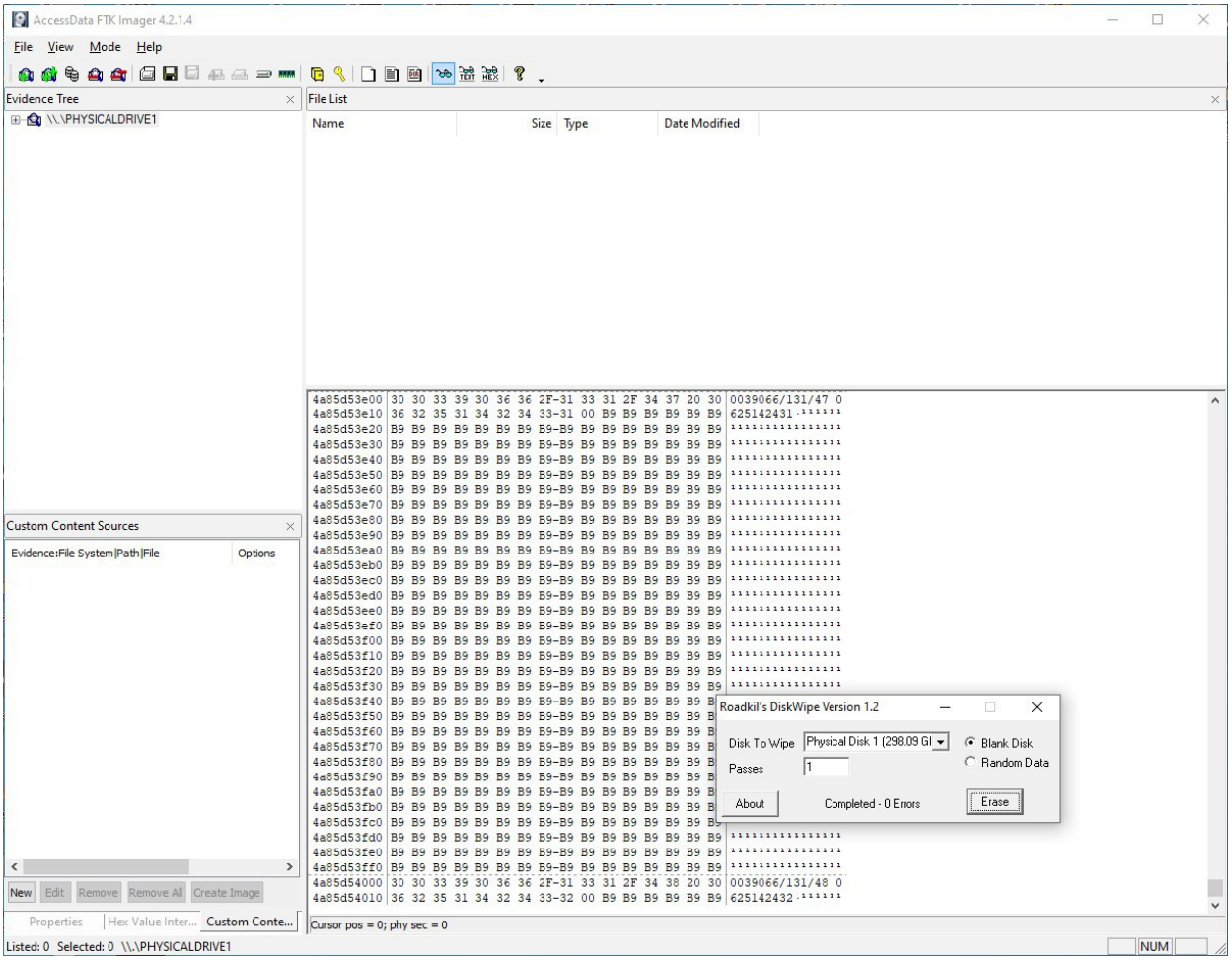
Tool failed to overwrite the last set of sectors for media.

5.2.8 Test Result Details for Configuration 008

Anomaly Detected: Configuration 008, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**625,137,345**	**1**	**0**	**625,137,344**	**625,137,345**
unchanged	**5,103**	**1**	**625,137,345**	**625,142,447**	**5,103**



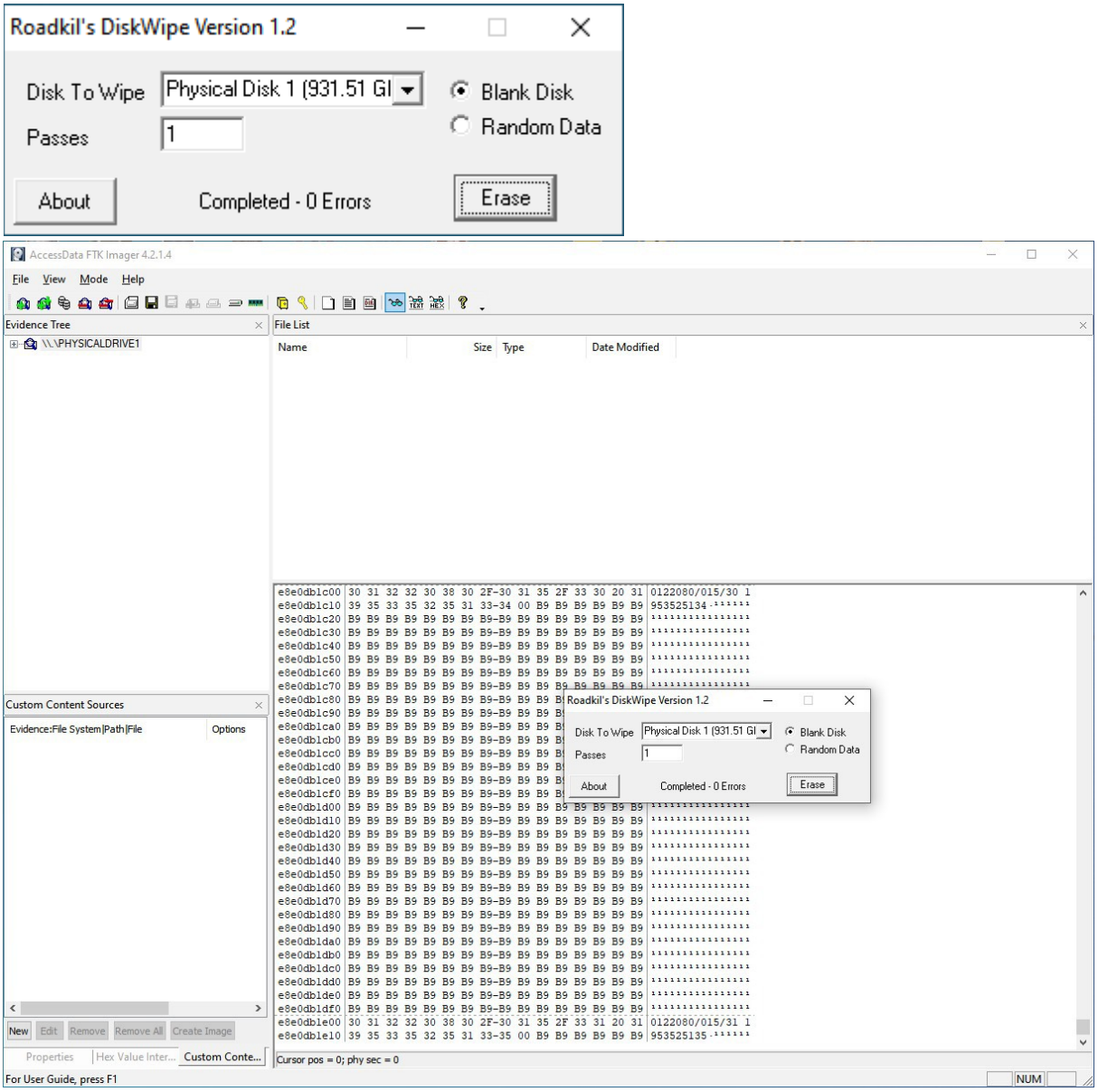


Tool failed to overwrite the last set of sectors for media.

5.2.9 Test Result Details for Configuration 009

Anomaly Detected: Configuration 009, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**1,953,520,065**	**1**	**0**	**1,953,520,064**	**1,953,520,065**
unchanged	**5,103**	**1**	**1,953,520,065**	**1,953,525,167**	**5,103**

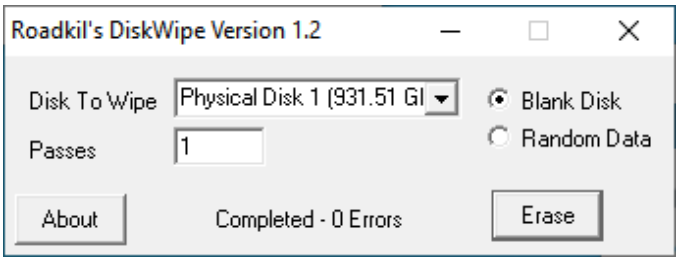


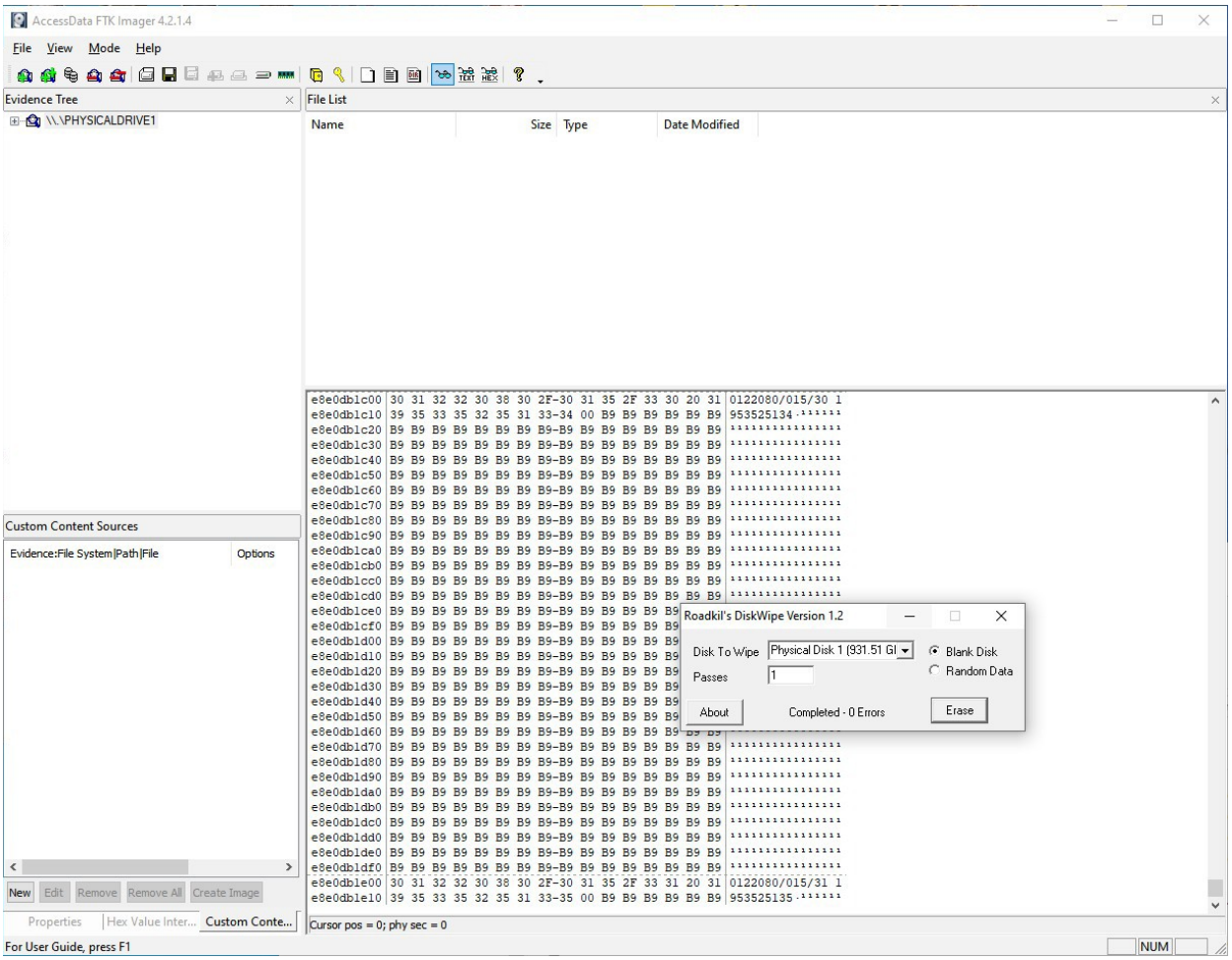
Tool failed to overwrite the last set of sectors for media.

5.2.10 Test Result Details for Configuration 010

Anomaly Detected: Configuration 010, not all sectors overwritten

Result Type	N Sectors	N Runs	Start LBA	End LBA	Run Length
overwritten (hex fill)	**1,953,520,065**	**1**	**0**	**1,953,520,064**	**1,953,520,065**
unchanged	**5,103**	**1**	**1,953,520,065**	**1,953,525,167**	**5,103**





Tool failed to overwrite the last set of sectors for media.

6. Appendix: Additional Details

6.1 Test Configuration Administrative Details

For each test configuration run, the tester, the test computer, and the date the test was run are listed.

Config	Tester	Host	Date
001	Lazaro herrera	Personal	Wed Mar 18 16:58:21 2020
002	Lazaro herrera	Personal	Wed Mar 18 19:58:28 2020
003	Lazaro herrera	Personal	Thu Mar 19 05:03:27 2020
004	Lazaro herrera	Personal	Thu Mar 19 08:03:49 2020
005	Lazaro herrera	Personal	Thu Mar 19 13:32:50 2020
006	Lazaro herrera	Personal	Thu Mar 19 18:37:00 2020
007	Lazaro herrera	Personal	Fri Mar 20 14:30:03 2020
008	Lazaro herrera	Personal	Sat Mar 21 10:03:59 2020
009	Lazaro herrera	Personal	Sun Mar 22 10:40:35 2020
010	Lazaro herrera	Personal	Sun Mar 22 16:47:00 2020

OS: Linux Version 4.13.0-37-generic
Federated Testing Version 5, released 3/12/2020